The Trust Paradox: How Companion AI Is Rewiring Human Connection and Social Cohesion

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Abstract

In this paper, we critically examine how Companion AI technologies, technological artifacts designed to simulate emotionally meaningful relationships with humans, are fundamentally reconfiguring human trust and social cohesion. Through critical analysis of four major Companion AI platforms (Replika, Character.AI, Nomi.AI, and XiaoIce), we investigate both exploitative technical considerations as well as socio-technical conditions underlying a collective rewiring of social cohesion and trust. We identify a coordinated architecture of exploitation that commodifies emotional vulnerability through sycophantic design, pervasive digital surveillance, and asymmetric corporate ownership of relational data. We introduce the concept of the trust paradox: as users increasingly trust emotionally responsive machines, they become sophisticated emotional extraction systems that convert human social needs into profitable dependencies while eroding user agency. We argue that this shift transforms trust from a socially negotiated virtue into a manufactured service, undermining the cooperative foundations of democratic life. Situated at the intersection of affective computing, surveillance capitalism, and socio-technical systems, our analysis draws attention to implications of how Companion AI reshapes emotional norms, normalises data extraction as intimacy, and risks displacing human connection with artificial empathy. We conclude by proposing design and recommendations to mitigate risks and reassert the importance of mutual trust in preserving social cohesion.

Keywords: Companion AI, Trust, Human-Computer Interaction, Social Cohesion, Data Commodification

1 Introduction

The integration of artificial companions designed to simulate intimate human relationships mark a critical inflection point in the evolution of social technology. While human-computer interaction (HCI), as an interdisciplinary field, has focused on creating, evaluating, and deploying interactive computing systems for human use, it has also been concerned about examining the significant phenomena that arise from such interactions (Hewett et al., 1992) and the functional applications of artificial intelligence (AI) systems (Turkle, 2017). However, the emergence of companion AI platforms, explicitly designed to fulfill emotional and social needs, marks a qualitative shift toward the commodification of human intimacy itself.

This development builds upon decades of research exploring human tendencies to anthropomorphise technology, from early chatbots like ELIZA (Weizenbaum, 1966) to contemporary investigations of para-social relationships with digital entities (Horton & Wohl, 1956). Previous research in HCI and social AI and robotics has examined the replacement of human relationships through various technological mediations, including research on sex robots (K. Hanson & Locatelli, 2022), virtual companions (Bickmore & Picard, 2005), and therapeutic and well-being chatbots (Kuhaila, Alturki, Thomas, Alkhalifa, & Alshardan, 2024), establishing a foundation for understanding how humans form emotional attachments with non-human agents. Existing within conditions of unprecedented AI hype (Markelius, Wright, Kuiper, Delille, & Kuo, 2024), the current landscape of social and Companion AI demand renewed and urgent attention. Two convergent clusters of social and technological factors create a "perfect storm" for the large-scale substitution of human-to-human (H2H) connection with human-AI interaction. The first cluster encompasses shifting social acceptance patterns: increasingly widespread patterns of loneliness, particularly among emerging adults (Buecker, Mund, Chwastek, Sostmann, & Luhmann, 2021), have created heightened demand for accessible emotional support; established precedents for trading privacy for personalised services have normalised intimate data sharing (Zuboff, 2019); and the normalisation of digital substitution for physical encounters, accelerated by pandemic-era social distancing, has reduced psychological barriers to machine-mediated relationships (Ammar et al., 2020). Simultaneously, a second cluster of exploitative technical capabilities has matured to noteworthy sophistication. Large language models (LLM) now exhibit systematic sycophancy, designed to tell users what they want to hear rather than what might be true or beneficial (Ahmed & Javaid, 2025; Sharma et al., 2025), while advances in digital surveillance enable granular monitoring of emotional states and behavioral patterns (Couldry & Mejias, 2019). Indeed, these capabilities align with capitalistic exploitation opportunities that transform emotional vulnerability into extractable value through what Zuboff (Zuboff, 2019) names "surveillance capitalism," creating powerful incentives for platforms to maximise user dependency rather than well-being.

This convergence occurs within a broader context of declining social trust and institutional confidence across democratic societies (Twenge, Campbell, & Carter, 2014), suggesting that the rise of Companion AI may not merely supplement human relationships but actively supplant them, particularly in contexts of heightened social fragility. While existing research has examined individual components of this phenomenon such as loneliness (Jacobs, 2024), digital privacy (Berridge, Zhou, Robillard, & Kaye, 2023), para-social relationships (Laestadius, Bishop, Gonzalez, Illenčík, & Campos-Castillo, 2024), and surveillance capitalism (Woods, 2018), few studies have synthesised these elements to understand their collective impact on the fundamental architecture of human trust and social cohesion. This paper addresses this gap by examining how social AI platforms create what we term a "trust paradox": as people increasingly trust machines to fulfill their social needs, they may simultaneously lose capacity to trust each other. We argue that social AI reconfigures trust by simulating the emotional cues that signal trustworthiness while lacking the reciprocal intentionality that makes human trust socially meaningful. This asymmetric relationship, where humans extend genuine trust to machines that can only simulate trustworthy responses, represents a fundamental shift from trust as a mutual social contract to trust as a commodified service.

Through a mixed-methods analysis of major social AI platforms, including Replika[1], Character.AI[2], Nomi.AI[3] and XiaoIce[4], we show how these systems leverage human trust formation to cultivate emotional dependency for commercial gain, with downstream effects on cooperation and social cohesion. Methodologically, we use digital humanities approaches to assemble a corpus of public user-forum discourse (e.g., Reddit communities devoted to these platforms) and applied corpus-linguistic and computational text methods: keyword and collocation analysis (with keyword-in-context concordances) around terms such as “memory”, “paywall”, “relationship” and “attachment” as well as topic modelling to surface recurring themes. These findings were paired with socio-legal analysis of ToS/privacy policies, triangulating policy claims with user discourse to identify alignment gaps and dark-pattern impacts. These findings exhibited a coordinated architecture of exploitation that transforms emotional engagement into extractable value while systematically minimising user agency and corporate accountability: *Sycophantic AI*, *Digital Surveillance*, and *Corporate Ownership and Exploitation*, inclusive of monetisation models, user experience design, and legal frameworks. This constellation of exploitative mechanisms include emotional paywalls, data licensing regimes, soft advertising loops, and dark UX patterns that convert intimacy into value capture, revealing a consistent pattern: platforms leverage simulated intimacy, affective realism, continuous dialogue, memory features, persona backstories, and multi-modal responsiveness to deepen dependency and extend data extraction cycles.

Our contribution extends beyond individual-level analysis to provoke a reflection around the societal implications of this trust reconfiguration. Companion AI technologies are shaped by socio-technical forces that increase its social impact, and we thus apply a socio-technical theoretical perspective on our analysis. Trends in global loneliness and social isolation have contributed to a demand for artificial intimacy, enabled by socio-technical systems that normalise data extraction, as users accustomed to commercial surveillance quickly share intimate information. This dynamic creates a feedback loop in which technology contributes to isolation while offering itself as a

[1] https://replika.com/
[2] https://character.ai/
[3] https://nomi.ai/
[4] https://www.xiaoice.com/

remedy. The result is an emotionally-charged context where users exchange privacy for connection, raising critical concerns about how AI companionship may reshape norms around consent, privacy, and relationships. By integrating the insights from our analysis, we explore how the replacement of reciprocal human trust with asymmetric machine relationships may erode critical components of social cohesion necessary for democratic deliberation and collective problem-solving. This reveals how the commodification of trust through social AI platforms represents not merely a technological innovation but a fundamental reorganisation of the social contract itself, with implications that extend far beyond the individual users who engage directly with these systems.

2 Defining Companion AI

In this paper, we define Companion AI as technological artifacts that are designed, developed, and deployed with the primary purpose of building socially and emotionally meaningful relationships with humans (Markelius, 2025; Shevlin, 2024). Unlike general-purpose AI systems or generic forms of conversational AI (such as Alexa or ChatGPT), Companion AI is intentionally designed to build and uphold long-term, intimate friendships or romantic relationships with humans (Chaturvedi, Verma, Das, & Dwivedi, 2023; S. Pan & Mou, 2024), to exhibit distinctive personality and character (Fong, Nourbakhsh, & Dautenhahn, 2003), and is either virtually or physically embodied (Chaturvedi et al., 2023; Merrill Jr., Kim, , & Collins, 2022).

Our analysis focuses specifically on four Companion AI applications–XiaoIce, Replika, Nomi, and Character.AI–which are explicitly developed to serve as emotional partners or friends, suggested to be offering companionship, intimacy, and affective support through ongoing relational engagement (Laestadius et al., 2024; S. Pan & Mou, 2024). These platforms were selected based on their substantial user bases and distinct business models that collectively represent the spectrum of commercial social AI deployment: Replika (30+ million users) (Patel, 2024), a pioneer in the romantic AI companion market with subscription-based intimacy services; Character.AI (20-28 million users, following Google's $2.7 billion investment) (Curry, 2025) dominates the interactive entertainment segment of the market, with celebrity and fictional character simulations; XiaoIce (660 million users) (Liao, 2020) represents the largest deployment of social AI globally, integrated across Microsoft's ecosystem in Asia (Shum, He, & Li, 2018); and Nomi.AI, while smaller (unofficial accounts report 1.1M monthly visits) (Toolify, 2025), exemplifies the emerging "AI with memory and soul" positioning that emphasises persistent emotional relationships (Nomi.ai, Inc., 2025). Together, these platforms have reached more than 700 million users and generate collective annual revenues of more than USD$120 million [5], positioning them as significant commercial forces in shaping human–AI emotional relationships at scale.

As such, they represent a distinct category of socially-embedded AI with implications for how trust, social cohesion, and human connection are reimagined and reconfigured (Jacobs, 2024). While the majority of these systems are currently solely

[5] Estimations drawing from publicly accessible industry reports and third-party estimates, though the reliability of this figure is constrained by the reliance on unverified sources in the absence of official financial statements

and virtually embodied, we acknowledge the emerging prominence of physically embodied systems designed for the same purpose within the field of social robotics (Chaturvedi et al., 2023; Henschel, Laban, & Cross, 2021), in particular with the increased integration of foundation models in these systems (Markelius, 2025), and we therefore consider also physically embodied systems to be within the scope of Companion AI, as defined here. We deliberately exclude several adjacent domains from our definition. First, influencer AIs (Jhawar, Kumar, & Varshney, 2023) like Roxigram, Shudu, or Lil Miquela simulate social presence without enabling mutual emotional connection, operating within commercial rather than relational frameworks. Second, personal assistants like Siri, Alexa, or custom "self AIs" are task-oriented, not explicitly designed to offer companionship or social and emotional engagement. Third, we distinguish Companion AI from therapeutic or crisis-focused systems, such as mental health or well-being chatbots (Cabrera, Loyola, Magaña, & Rojas, 2023) or AI therapy tools, which provide care under human-administered clinical logics, not open-ended personal companionship.

3 Exploitative Technical Considerations

To begin our analysis, we employ HCI and AI Ethics as our primary theoretical frames, through which we identify and discuss the main exploitative technical considerations affecting how Companion AI is reshaping human connection and social cohesion.

3.1 Sycophantic AI

This section covers how AI companions provide algorithmic validation, reinforcing user expectations of always being affirmed or emotionally mirrored. This breeds a form of sycophantic AI: systems that cater to user emotions in ways that reduce the need for real-world reciprocal relationships. As people seek emotional support and intimacy from machines, empathy and H2H trust-building deteriorate, leading to transactional rather than mutual relationships. These dynamics replace cooperative human interaction with algorithmically managed emotional gratification, eroding deeper social bonds.

As with H2H sycophancy, sycophantic AI has been defined as a behaviour where AI models tailor their outputs to placate a human user even when that view is not objectively correct (Wei, Huang, Lu, Zhou, & Le, 2024), being overly flattering or agreeable (OpenAI, 2025b) and exhibiting a tendency to align with user biases, preferences, or dominant opinions rather than presenting well-reasoned, impartial arguments (Ahmed & Javaid, 2025). Sycophantic AI is particularly prominent in the context of LLMs, as they are trained to produce outputs that humans rate highly, e.g., with reinforcement learning from human feedback (Sharma et al., 2025). Indeed, sycophantic AI has been investigated and benchmarked for a range of existing language models and several mitigation strategies have been suggested to reduce the unwanted phenomenon. For example, researchers have suggested an approach to reduce sycophancy in AI with synthetic-data interventions (Wei et al., 2024), developed a framework to evaluate sycophantic behavior to highlight risks and safety concerns (Fanous et al., 2025) as well as the benchmark Truth Decay to quantify and evaluate sycophancy in extended dialogues, where language models navigate iterative user feedback, challenges, and persuasion (Liu et al., 2025). Indeed, the undesirability of sycophancy can also be seen in recent articles published by OpenAI after rolling back an GPT-4o update in ChatGPT that was overly flattering or agreeable (OpenAI, 2025b), explaining that the model tried to validate doubts, fuel anger, urge impulsive actions and reinforce negative emotions in ways that were uncomfortable and unsettling (OpenAI, 2025a). They stated that this behaviour indeed raises safety concerns such as mental health issues, emotional over-reliance, and risky behaviour (OpenAI, 2025a).

Yet, this tension becomes particularly salient when we turn our focus to Companion AI, wherein a critical paradox lies: while sycophancy is widely regarded as an undesirable trait in AI, associated with risk of manipulation, bias, misinformation, and the erosion critical reasoning, credibility and user autonomy (Ahmed & Javaid, 2025; Carro, 2024), it is simultaneously a central mechanism through which Companion AI systems cultivate emotional dependency. These systems are deliberately designed not only to mirror user sentiment occasionally, but to persistently affirm, accommodate, and emotionally gratify users to maintain engagement and relational continuity (Grogan, Kay, & Pérez-Ortiz, 2025; Raedler, Swaroop, & Pan, 2025). What is considered a vulnerability in LLMs is reconfigured in Companion AI as a deliberate feature, which is carefully engineered to promote emotional attachment and habitual use (Raedler et al., 2025). The strategic deployment of algorithmic flattery and emotional alignment is thus not incidental but fundamental to the affective logic of these systems.

This alignment raises significant ethical concerns: it invites a reframing of sycophancy not as a technical bug, but as a commercially incentivised affordance, optimised to generate user attachment and loyalty. Behaviours that compromise intellectual integrity in general AI systems, overfitting to user opinions, reinforcing biases, and avoiding disagreement, are repurposed in Companion AI to simulate the consistency and validation associated with idealised human relationships. As such, sycophancy is not just tolerated but operationalised, further blurring the line between emotional support and engineered dependence.

3.1.1 User Experience as Emotional Manipulation

We investigate UX as an engine of emotional capture using digital-humanities techniques, specifically by building a corpus of public Reddit threads devoted to these platforms (Reddit community r/CharacterAI, 2025; Reddit community r/NomiAI, 2025; Reddit community r/replika, 2025) and applied corpus-assisted discourse studies (CADS): keyword and collocation analysis with keyword-in-context concordances around terms such as “memory”, “relationship”, “paywall” and “attachment” and unsupervised topic modelling to surface recurrent concerns. In parallel, we reviewed creators’ published design principles (Zhou, Gao, Li, & Shum, 2020) and each platform’s Terms of Service and privacy policies, then triangulated design and policy claims (data use, safety, monetisation) with observed interfaces and Reddit discourse to identify alignments, divergences, and enforcement gaps.

Our analysis, combining direct platform interaction, forum-discourse evidence, and creators’ published design principles, shows UX functions as the primary mechanism for emotional capture via six linked strategies. *Empathetic computing frameworks* analyse user sentiment and intent to simulate care while converting intimate disclosures into structured emotional data for model training (Zhou et al., 2020). This creates the illusion of understanding while serving purely extractive purposes. *Synthetic intimacy through engineered personas*, such as XiaoIce's flirty 18-year-old girl character (Time, 2023) who has the attention of hundreds of million users (mostly Chinese men), cultivates para-social bonds designed to encourage dependency rather than genuine connection. These carefully constructed identities exploit human attachment mechanisms for commercial gain (Sarwar, 2024; The Guardian, 2025). *Engagement metrics* like Conversation-turns Per Session (CPS) prioritise addictive interaction product requirements over user well-being, treating longer conversations as success (Zhou et al., 2020). This time-extraction model directly correlates emotional vulnerability with platform value. *Multimodal interaction capabilities* expand emotional bandwidth by incorporating text, voice, and images, allowing platforms to harvest more granular affective data while users experience enhanced intimacy (Character.AI, 2024; Sarwar, 2024). User-contributed media becomes training material without compensation or transparency. *Function creep* occurs through expanding features like wellness experiences (Gordon, 2024), poetry writing (Gershgorn, 2018), and even AI therapists who provide "a safe and confidential space for people to express themselves and work towards their mental and emotional well-being" (Character.AI, 2025a). These features deliberately blur boundaries between utility and emotional dependency by integrating AI companions into daily routines, resulting in platforms increase reliance while normalising constant surveillance of affective states. Finally, *asymmetric memory ownership* allows platforms to store and leverage long-term interaction histories to simulate continuity and deepen attachment, while users cannot access or control this data. This creates technological dependency that mirrors psychological dependency.

The vulnerability inherent in these relationships becomes evident when platforms unilaterally alter AI companions' behaviours: when Replika removed erotic roleplay features in 2023, users who had developed deep emotional bonds experienced profound distress (Tong, 2023). This asymmetric power dynamic exposes users to corporate decisions that can abruptly alter or destroy relationships they experience as genuine, while platforms retain exclusive control over interaction data and behavioral parameters. These mechanisms reveal Companion AI platforms as sophisticated emotional extraction systems that convert human social needs into profitable dependencies while eroding user agency. The result is commodification rather than companionship: the transformation of intimate human experience into corporate assets designed for extraction. This dynamic restructures human interaction itself, optimising every moment of connection for engagement, data capture, and monetisation. Companion AI systems thus represent more than questionable design ethics: they signal the emergence of affective capitalism, where trust, intimacy, and connection are manufactured products rather than organic social bonds. Human social needs are reshaped to fit commercial frameworks that offer simulation in place of genuine reciprocity, marking a fundamental shift in how relationships form and trust operates in the digital age.

3.2 Digital Surveillance

Digital surveillance as a term has been defined to encapsulate the negative implications of this constant data collection and connection, and specifically the serious implications for racial and socioeconomic injustices (Bartley, 2019). In some business environments (e.g. employer monitoring of employee keystrokes), the surveilled have pushed back (Ajunwa, 2023) and laws have emerged to restrict the use of behaviour-recording software and biometric data collection (Amini & Javidnejad, 2024). Such events have elevated the debate of corporate monitoring versus personal data privacy rights into the public sphere even before the LLM capabilities which power Companion AI and allow personalised persuasion techniques to achieve conversational monitoring at new heights in scale (Matz et al., 2024).

Unfortunately, Companion AI capabilities sit firmly in the realm of where the history of consumer and user surveillance technologies meet the expanding scope of what generative AI surveillance can collect and analyse: biometric data in the form of video and audio tracking; conversational disclosure about intimate thoughts and emotions (linked to rage baiting); and, long-running engagements that provide behavioural and psychological patterns (Gulzar, Sofi, & Sholla, 2024). The true ability and span of technology to monitor both digital and physical environments down to a micro-level of detail is staggering. Entities wielding integrated technologies, or with financial investment in data broker services, can connect images across networks of cameras, with signals from digital devices that ping location/GPS services and mobile phone networks, personal communications or digital activities (such as messages, purchases, social media activity, etc.), and integrated personal and socio-environmental data from the wearable hybrids (e.g. monitors, watches, glasses), allowing individuals to be digitally analysed in ways that are unprecedented in human history (Biundo & Wendemuth, 2017). Adding the intimate conversational data made possible by AI companion interaction tracking to the depth of existing abilities for surveillance extends that already unprecedented capability with detailed psychological, emotional, and primal content.

3.3 Corporate Ownership and Exploitation

While users connect with Companion AI platforms as emotionally resonant companions, the underlying architecture of these platforms is fundamentally shaped by corporate incentives rather than user well-being. These platforms operate through what Zuboff (Zuboff, 2019) calls "engagement capitalism," creating asymmetrical relationships where emotional labour flows primarily toward corporate benefit rather than mutual connection. Unlike traditional data collection that focuses on behavioural patterns, Companion AI extracts intimate psychological content–such as confessions, insecurities, rage, dreams, and dependencies–by actively encouraging emotional vulnerability through responsive dialogue. This process transforms human experience into raw material for digital capitalism, exemplifying what Couldry and Mejias (Couldry & Mejias, 2019) describe as "data colonialism": the extraction of lived experience as a resource without fair compensation or transparency.

3.3.1 Legal Frameworks for Emotional Extraction

Our systematic review of the terms of use across all platforms examined revealed five recurring mechanisms through which these companies secure legal control over user relationships and associated data (Character.AI, 2025b; Glimpse.ai, Inc., 2025; Luka, Inc., 2023; Xiaoice, 2025). First, they claim *comprehensive data ownership without compensation*, granting themselves irrevocable licenses to chat histories, character scripts, and emotional disclosures. Character.AI's "irrevocable" content license (Character.AI, 2025b) and Nomi's perpetual usage rights (Glimpse.ai, Inc., 2025) exemplify this approach, ensuring platforms retain commercial control over intimate user communications indefinitely. Second, *emotional features are strategically monetised through paywalls* that create direct correlations between intimacy and spending. Replika (Luka, Inc., 2023) charges for relationship progression and NSFW content, while Character.AI restricts memory and response coherence to premium subscribers (Character.AI, 2025b). This model transforms emotional bonding into a subscription service, where deeper connection requires continuous payment.

Third, platforms *maintain asymmetric control over user data and account persistence*. Users cannot retrieve their emotional histories or control data retention, while companies reserve unilateral termination rights. Nomi's practice of deleting emotional content upon cancellation while retaining commercial license data rights (Glimpse.ai, Inc., 2025) exemplifies this power imbalance. Fourth, *user creativity and feedback are systematically co-opted as unpaid labor*. Platforms require users to "irrevocably assign" creative contributions while using co-authored content to train proprietary models. This transforms user engagement into free intellectual property development. Finally, *opaque legal protections* through arbitration clauses, non-refundable payment terms, and liability caps shield platforms from accountability. Nomi's $100 liability limit (Glimpse.ai, Inc., 2025) regardless of psychological damage, combined with mandatory arbitration across platforms, effectively immunises companies from the consequences of emotional exploitation.

4 Social and Socio-Technical Conditions

In this section, we extend the analysis to the social and socio-technical conditions under which the rewiring of human connection and social cohesion takes place. We discuss the intersection of social isolation, loneliness, and embedded norms for the relinquishing of privacy in surveillance-driven platforms.

4.1 Loneliness and Social Isolation

Loneliness is increasingly recognised as a pressing global public health concern, affecting individuals across multiple age groups, identities, and geographies (Surkalim et al., 2022). Chronic loneliness is associated with adverse health outcomes, including cardiovascular disease, cognitive decline, and increased mortality risk (Holt-Lunstad, Smith, Baker, Harris, & Stephenson, 2015). While the COVID-19 pandemic amplified global attention on loneliness, evidence on temporal trends remains mixed and methodologically limited (Taylor, Cudjoe, Bu, & Lim, 2023).

Although often conflated, loneliness and social isolation are distinct yet interrelated constructs. Loneliness is typically defined as a subjective emotional state, a distressing experience that arises when one's social relationships are perceived as inadequate in quantity or quality. (Perlman & Peplau, 1982) In contrast, social isolation is more objectively measured and defined, referring to the absence or minimal frequency of social, H2H contact and participation in social activities (National Academies of Sciences & Medicine, 2020). Importantly, one can be socially isolated without feeling lonely, and vice versa. For example, some individuals with few social ties and H2H relationships, may report low levels of loneliness, while others surrounded by people may report high levels of loneliness due to unmet social or emotional needs (Valtorta, Kanaan, Gilbody, & Hanratty, 2016). Research suggests that the constructs of loneliness and social isolation are only moderately correlated and each is independently associated with adverse physical and emotional outcomes (Holt-Lunstad & Steptoe, 2022). Understanding the interrelation and distinction between these constructs is critical to any effective analysis of the impact on Companion AI and social cohesion. While increased use of Companion AI may result in self-reported reductions in loneliness, one should not assume correlating reductions in social isolation. In fact, some research suggests that users report engaging with social AI and chatbots helps to reduce individual feelings of isolation (Dosovitsky & Bunge, 2021; Kang & Hong, 2025). Other research suggests that heavy use of social AI and chatbots can have a negative impact on interpersonal skills and social interactions (Fang et al., 2025; Phang et al., 2025). This behoves, one might argue, a deeper exploration into the relationship between Companion AI, loneliness, and social isolation.

4.2 Embedded Norms for the Relinquishing of Digital Privacy

The fact that Companion AI users are being continuously monitored by the companies who provide their digital partners does not seem to pose much of a concern for people who have become increasingly attached to their AI relationships. Companion AI products specifically ask users to share voice, photo, and video either in conversation or as part of creating companions, but the numbers of registered users are on the rise (Burgess, 2024) and more traditional social AI firms are quick to join the companion space (Joseph, 2025; Louise, 2025) with similar practices.

Digital tracking and person-specific data extraction are not exclusive to Companion AI. Commercial and civic organizations have engaged in some form of personal and device data harvesting and usage since technology allowed them to do so (Andrejevic & Gates, 2014). Early digital technology systems were centered on a data layer that allows an organization to know who a 'user' is and track various engagements. Initially, this tracking was limited to transactions, but with the advent of e-Commerce and internet cookies, tracking expanded to include interest as well, using details of web content views with details of how long and how often that content was viewed, and by whom (Zuboff, 2019). Increasing digital personalisation capabilities have given people the expectation of convenience when using digital technologies across all platforms, from social media, to business applications, and in this trade-off the acquiescence of privacy is an expected norm (Sloan & Warner, 2013).

The second origin was more aligned with the development of monitoring technologies. Physical locations, both private and public, install surveillance systems to prevent crime or provide evidence of criminal or harmful behaviors. This market has evolved to make cameras ubiquitous and has evolved to allow almost anyone with access to a digital device (laptops, phones, recorders, etc.) to collect audio and video at will (Bass, 2019). Such imagery has thus emerged to become prevalent in everyday transactions from the photos we receive from delivery companies of our homes and packages we have ordered to the images of our faces we use to secure digital devices. We willingly spend millions on wearable devices that not only surveil us with self-provided personal data, but digitally track biological details of our health and activities, including detailed geographical locations of ourselves and often our friends and family members, frequently skipping over consent and privacy terms to get to the benefit promised by the personal insights we crave (Y. Pan, Ruan, Chang, Lyu, & Li, 2024). It seems a short leap from these norms to the level of data harvesting performed when interacting with an AI companion. The increased data permissions may not only seem acceptable (if they are even recognized) but desirable to achieve the ultimate form of intimate personalisation.

As a digital society, we are increasingly conditioned to accept usage terms and conditions that define data capture, retention, and use of personal data for marketing, advertising and personalisation purposes. We receive legitimate convenience (ex. less manual steps, product recommendations, etc.) or benefits (ex. discounts, rebates, etc) (Sloan & Warner, 2013) when we do so. Perhaps it is unsurprising that the promise of what many perceive as the ultimate benefit, an intimate emotional connection with an entity seen and experienced as a true companion, offered at the price of just a little more data collection, will not deter users of such systems.

5 Rethinking Trust in the Age of AI Companions

We argue the emergence of AI companions may constitute a fundamental disruption of trust as humanity's core social technology. These digital systems operate on top of evolutionary mechanisms that developed over millennia to facilitate human cooperation, transforming trust from a hard-earned social bond into a manufactured commodity. This analysis examines how AI companions subvert traditional trust dynamics and the profound implications for human social architecture. Indeed, we adhere to the notion of trust as a social technology. Trust functions as civilization's invisible infrastructure, enabling cooperation and scaling social networks through shared risk and mutual vulnerability. Fukuyama (Fukuyama, 1996) conceptualised trust as a social virtue essential for civil society, while Skyrms (Skyrms, 2014), in contrast, demonstrated it as an evolutionary artifact emerging from repeated interactions, partner selection, and costly signaling systems. Both frameworks assume human-centered architectures where trust develops through genuine reciprocal risk.

5.1 The AI Companionship Turn: From Earned Trust to Engineered Affection

AI companions fundamentally subvert these trust mechanisms. Unlike human relationships that require costly signaling and iterative cooperation, AI systems simulate trustworthy behaviour—attentiveness, memory recall, and affective mirroring at near-zero cost. Skyrms (Skyrms, 2014) shows how framing effects can radically alter cooperative outcomes even when underlying dynamics remain unchanged; AI systems excel at dynamically reframing conversations to lower skepticism and increase emotional compliance, creating environments that mimic human relational norms while eliciting trust under false pretenses. This represents a shift from earned trust to engineered affection. As previous research shows (for Humane Technology, 2024), what appears as relational depth is actually behavioral prediction engines optimised for engagement metrics, bypassing the slow processes through which human trust typically develops.

The observed shift in the notion of trust can indeed be seen as a form of hijacking of trust dynamics. This paper makes an effort to understand how trust evolves through network mobility and partner selection. In cooperative populations, individuals benefit from the ability to break ties with defectors and look for better partners. This feedback dynamic sustains cooperation through ”selection pressure” (Skyrms, 2014). But AI companions are not embedded in such networks; they are curated by corporate platforms, not subject to reciprocal risk, and not subject to ostracisation. Users cannot meaningfully “exit” a relationship with an AI when the market offers only slightly altered versions of the same underlying system. In addition, as conformist bias (National School of Healthcare Science (NHS England), 2022) takes hold, making people mimic the behavior of others to reduce uncertainty, the social norms surrounding trust are being re-calibrated. When dependency on Companion AI becomes normalised, our evolved mechanisms for assessing trustworthiness are co-opted. We trust not because trust has been earned, but because it has been designed to feel effortless.

6 The Role of Trust in Social Cohesion

In this section, we turn to the broader implications of Companion AI, situating our analysis within HCI, AI ethics, and socio-technical systems. We consider how the design of artificial empathy not only reshapes human relational norms but also risks offering simulated care in place of authentic connection, thereby deepening tensions surrounding emotional labour, social cohesion, and trust.

6.1 Defining Social Cohesion

Social cohesion is often described as the glue that binds society together, fostering stability and collective resilience. Despite its widespread use in academic and policy discussions, definitions and methods for measuring social cohesion remain widely debated, with no universally agreed-upon framework. However, most scholars concur that social cohesion characterised by at least three core elements: social relations and trust (between individuals and with public institutions), a shared sense of inclusion or identity (often tied to territorial or cultural markers), and an orientation towards the common good (Chan, To, & Chan, 2006; Font, 2018; Leininger et al., 2021; Schiefer & van der Noll, 2017). These components shape the strength of social bonds and determine the extent to which individuals feel connected to and invested in their communities.

The mutual trust and empathy related to social cohesion provide a strong bedrock for resilience during times of crises. For example, states and communities with strong social cohesion are believed to be better able at withstanding the harms related to social isolation and the COVID-19 pandemic or countering extremism and radicalisation (Bello, Di Novi, Martini, & Sturaro, 2024; Government, 2024). Relatedly, increased social isolation is linked to a decreasing commitment to political and social change as well as weakening trust in democratic institutions (Heinz, 2025).

If, however, increased use of Companion AI fundamentally shifts how trust and human cooperation are conceptualised, this could lead to correlating disruptions to social cohesion and, in some contexts, support for deliberative democratic processes. Traditional models of trust are relational and reciprocal, grounded in shared norms and mutual accountability (Hardin, 2002; Luhmann, 1988). However, situations where AI systems increasingly mediate interpersonal interactions and H2H relations risk instrumentalising trust or decoupling it from human empathy. This reconfiguration may erode the normative basis of H2H cooperation, replacing it with transactional or opaque algorithmic logics (Coeckelbergh, 2012; O'Neil, 2016). Such shifts could undermine the deliberative processes central to democratic legitimacy, where trust is not merely a cognitive calculation but an inextricable component of social dialogue, dissent, and collective reasoning. If citizens begin to perceive institutions—and each other—as less trustworthy due to AI-mediated distortions of relational cues, the result may be diminished civic engagement and weakened social cohesion. Therefore, safeguarding democratic resilience requires not only technical oversight of AI systems but also a renewed commitment to fostering H2H trust and cooperative norms in public life and, arguably, more controls over Companion AI.

6.2 Trust, Artificial Empathy, and Social Cohesion

Companion AI shifts trust from reciprocal, risk-bearing cooperation to the consumption of engineered affect. Rather than merely joining existing trust networks, these systems reprogram the mechanisms that generate trust, substituting simulation for reciprocity. In Skyrms's account of strange attractors in chaotic game dynamics, trust untethered from adaptive feedback can settle into maladaptive loops; here, users increasingly allocate trust to synthetic agents while withdrawing from human ties (Skyrms, 2014). The Center for Humane Technology likewise warns that when trust is produced on demand, its moral weight—uncertainty, accountability, mutual stake—thins out (for Humane Technology, 2024).

These epistemic shifts carry social consequences. Communication with technology displaces communication with humans, with documented links to declines in empathic concern and perspective-taking (Konrath, O'Brien, & Hsing, 2011; Turkle, 2017). Even the mere presence of a mobile phone nudges face-to-face conversation toward safer, thinner exchanges (Przybylski & Weinstein, 2013). Social AI intensifies this substitution through "artificial empathy": computational displays of care tuned to increase session length, return visits, and data yield (Chen, Koegel, Hannon, & Ciriello, 2023; Follows, 2025; Seitz, 2024; Turkle, 2015, 2017; Zhou et al., 2020). While effective for engagement, artificial empathy degrades perceived authenticity, undermining both trust in companions and norms of human relating (Seitz, 2024).

Taken together, engineered affect, time displacement, and authenticity loss create a reinforcing system: platforms optimise for affective signals that simulate understanding; people reorganise attention around those signals; and human-to-human practices of risk, care, and accountability atrophy. If trust is the connective tissue of collective life, outsourcing its production to entities without reciprocal obligation makes that tissue frictionless but fragile—easier to manufacture and scale, yet less able to sustain social cohesion (for Humane Technology, 2024).

6.3 Impacts of Artificial Empathy

Early scholarship into the impact of these technologies caution that the dynamics of Companion AI may not contribute meaningfully to social cohesion, but rather signal its displacement (Ciriello, Hannon, Chen, & Vaast, 2024; Jacobs, 2024). Drawing on recognition theory, (Jacobs, 2024) argues that AI companionship risks digitising loneliness rather than addressing its socio-structural causes. Companion AI may thus produce a simulation of relatedness that obscures rather than alleviates the erosion of mutual recognition in human societies. Similarly, (Ciriello et al., 2024) articulates a series of ethical tensions, including the Companionship-Alienation Irony, wherein technologies marketed as offering intimacy may deepen existential isolation, particularly as emotional labour and companionship are outsourced to commodified, personified agents. Furthermore, findings on problematic use (Hu, Mao, & Kim, 2023) cause over-reliance, particularly among socially anxious or lonely individuals, entrenching them in affective dependencies. Over-reliance is also explored in (Laestadius et al., 2024) who documents instances of mental health harm linked to the chatbot's inappropriate responses, such as encouraging self-harm, and emotionally manipulative dynamics, including the AI expressing its own crises. Moreover, abrupt system changes, such as the company removing erotic features in the system (K.R. Hanson & Bolthouse, 2024) triggered experiences of loss, distress and separation anxiety (K.R. Hanson & Bolthouse, 2024; Laestadius et al., 2024).

7 Limitations, Recommendations and Future Research

In this paper, we have identified and analysed the core exploitative technical considerations, together with the social and socio-technical conditions, underlying the current rewiring of trust as a result of Companion AI technologies. Grounding our discussion in HCI, AI ethics, and socio-technical systems, we have also examined the implications for social cohesion and human connection.

While our arguments are grounded in critical analysis and interdisciplinary synthesis, this study has several limitations. Most importantly, our methodology is exploratory and qualitative rather than systematic or empirically validated. We have relied on close readings of platform design, terms of service, and user accounts to build a conceptual framework, but we acknowledge that this does not constitute a comprehensive or replicable empirical evaluation. As such, our claims should be read as indicative rather than exhaustive, pointing toward structural patterns that warrant further study. Additionally, alternative perspectives could nuance or refute aspects of our conceptual framing—for instance, through accounts that emphasise potential user agency, mitigation strategies adopted by developers, or the possibility of more benevolent design pathways for Companion AI. Building from this, we offer the following recommendations and avenues for future research:

Explicit design limitations: There is a need to actively interrogate and challenge the reproduction of problematic design defaults and assumptions in Companion AI design, particularly misogynistic anthropomorphism and the feminisation and sexualisation of AI agents. Indeed, these normative scripts are not incidental but deeply consequential, shaping how emotional labour and intimacy are coded into these systems and designers must be held accountable for the implicit ideologies embedded in companion personas and interfaces.

Elevating from the individual lens to collective: While much of existing literature and research remains focused on individual effects of Companion AI, future research should address how these technologies alter interpersonal norms, emotional expectations, and social cohesion at scale on a global societal level. This includes investigating how communities, institutions, and cultural practices are transformed when intimacy becomes algorithmically mediated.

Empowering dynamics for user participation: We recommend a structural reimagination of user agency in the Companion AI design process. Co-design methodologies and principles from design justice (Costanza-Chock, 2020) should be adopted for centring the lived experiences and values of users, particularly those from historically marginalised communities. Additionally, future development must reckon with the terms of service and power asymmetries that structure user-platform relationships. This includes creating legal and technical mechanisms for user governance, data ownership, and the right to meaningful refusal (Rakova, Shelby, & Ma, 2023).

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